

# Function Blocks

# Function Blocks

| 逻辑指令                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 定时器                                                                                                                                                                                                                                                                                                                                                               | 脉冲及触发器                                                                                                                                                                                                                                                                                                                             | 比较指令                                                                                                                                                                                                                                                                                                                              |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [AND] AND operation<br>[NAND] NAND operation<br>[OR] OR operation<br>[NOR] NOR operation<br>[NOT] NOT operation<br>[XOR] Exclusive-OR operation<br>[ROL_DW] DWORD rotate left<br>[ROL_W] WORD rotate left<br>[ROR_DW] DWORD rotate right<br>[ROR_W] WORD rotate right<br>[SHL_DW] DWORD shift left<br>[SHL_W] WORD shift left<br>[SHR_DW] DWORD shift right<br>[SHR_W] WORD shift right<br>[WAND_DW] Double word AND operation<br>[WAND_W] Word AND operation<br>[WNAND_DW] Double word NAND operation<br>[WNAND_W] Word NAND operation<br>[WNOR_DW] Double word NOR operation<br>[WNOR_W] Word NOR operation<br>[WNOT_DW] Double word NOT operation<br>[WNOT_W] Word NOT operation<br>[WOR_DW] Double word OR operation<br>[WOR_W] Word OR operation<br>[WXOR_DW] Double word exclusive-OR operation<br>[WXOR_W] Word exclusive-OR operation | [SWI_DLY] Switch-on and off delay<br>[TMR_FNCT] Pulse and delay<br>[DLY_BO] Delay for Boolean<br>[DLY_I] Delay for integer value<br>[DYL_R] Delay for Real value<br>[STPWATCH] Stop watch<br>[CYCLEGEN] Cycle generator<br><br><b>计数器</b><br>[CNT_DN] Down counter<br>[CNT_UP] Upward counter<br>[CNT_UPDN] Up and down counter<br>[CNT_BO] 8-way Boolean counter | [R_TRIG] Recognize positive edge and trigger pulse<br>[F_TRIG] Recognize negative edge and trigger pulse<br>[JK_FF] JK flip-flop<br>[SR_FF] SR flip-flop, set dominant<br>[RS_FF] RS flip-flop, reset dominant<br>[SR_FF_FB] SR Flipflop (function block), set dominant<br>[RS_FF_FB] RS Flipflop (function block), reset dominant | [CMP_I] Comparator for INT values<br>[CMP_DI] Comparator for DINT values<br>[CMP_R] Comparison of real values<br>[CMP_MS] Compare Multistate<br>[CMP_RNG] Comparison multistate range<br>[CMP_T] Comparator for TIME values<br>[CMP_DA] Compare date<br>[CMP_DATI] Compare date and time of day<br>[CMPTIDAY] Compare time of day |

# Function Blocks

| 数学运算指令                                       | 数学运算指令                                  | 数学运算指令                                                |
|----------------------------------------------|-----------------------------------------|-------------------------------------------------------|
| [ADD_DI] Addition double integer values      | [MAX_I] Maximum of INT values           | [AVE_TIME] Time-based average                         |
| [ADD_DW] Addition double word values         | [MAX_DI] Maximum of DINT values         | [AVERAGE] Average value of 8 real values              |
| [ADD_I] Addition Integer values              | [MAX8_R] Maximum of 8 real values       | [CHAR_LIN] Linear characteristic curve                |
| [ADD_R] Addition real values                 | [MIN_I] Minimum of INT values           | [CHARFNCT] Characteristic function with interpolation |
| [ADD_TI] Addition time                       | [MIN_DI] Minimum of DINT values         | [FORMULA] Formula                                     |
| [SUB_I] Subtraction integer values           | [MIN8_R] Minimum of 8 real values       | [INTEGRAL] Integral function                          |
| [SUB_DI] Subtraction double integer values   | [ABS_I] Absolute value of INT values    | [LGRTM] Logarithm functions                           |
| [SUB_DW] Subtraction double word values      | [ABS_DI] Absolute value of DINT values  | [POW_FNCT] Power functions                            |
| [SUB_R] Subtract real values                 | [ABS_R] Absolute value of REAL values   | [POW_XY] Power function to base X                     |
| [SUB_TI] Subtract time                       | [MOD_I] Modulo function of INT values   | [TGNMFNCT] Trigonometric functions                    |
| [MUL_I] Multiplication of integer values     | [MOD_DI] Modulo function of DINT values | [EPS_DI] Accuracy approximation of DINT values        |
| [MUL_DI] Multiplication double integer value | [LIM_DI] Limiter for DINT values        | [EPS_I] Accuracy approximation of INT values          |
| [MUL_R] Multiplication of real values        | [LIM_I] Limiter for INT values          | [HOLD_B] Hold binary value                            |
| [MUL_TI_R] Multiply time by real             | [LIMIT_R] Limit value supervision Real  | [HOLD_DI] Hold Double Integer value                   |
| [DIV_DI] Division of DINT values             |                                         | [HOLD_R] Hold real value                              |
| [DIV_I] Division of INT values               |                                         | [HOLD_USG] Hold unsigned value                        |
| [DIV_R] Division real values                 |                                         | [RANDOM] Random value generation                      |
| [DIV_TI_R] Division for time by real         |                                         | [RATE_LIM] Rate limiter                               |

# Function Blocks

| 数据转换指令                                                    |
|-----------------------------------------------------------|
| [BIT_I] Conversion bit pattern to integer values          |
| [I_BIT] Conversion integer values to bit pattern          |
| [BO_BY] BOOL to BYTE conversion                           |
| [BO_DW] BOOL to DWORD conversion                          |
| [BO_W] BOOL to WORD conversion                            |
| [BY_BO] BYTE to BOOL conversion                           |
| [BY_DW] BYTE to DWORD conversion                          |
| [BY_W] BYTE to WORD conversion                            |
| [DI_DW] DINT to DWORD conversion                          |
| [DI_I] DINT to INT conversion                             |
| [DI_R] DINT to REAL conversion                            |
| [DI_TI] Conversion double integer value to time           |
| [R_I] Conversion of real values to integer values         |
| [R_DI] Conversion of real values to double integer values |
| [W_I] WORD to INT conversion                              |
| [W_DW] WORD to DWORD conversion                           |
| [DW_BO] DWORD to BOOL conversion                          |
| [DW_DI] DWORD to DINT conversion                          |
| [DW_R] DWORD to REAL conversion                           |
| [DW_TI] Converting Double Word to Time-Signal             |
| [DW_TIDAY] Conversion double word value to time of day    |

| 数据转换指令                                                  |
|---------------------------------------------------------|
| [DW_W] DWORD to WORD conversion                         |
| [I_DI] INT to DINT conversion                           |
| [I_DW] INT to DWORD conversion                          |
| [I_R] INT to REAL conversion                            |
| [I_W] INT to WORD conversion                            |
| [I_LOGI] Conversion of integer value to logical signals |
| [LOGI_I] Conversion logic signals to integer value      |
| [MS_MS] Conversion multistate to multistate             |
| [NEG_I] Inverter for INT values                         |
| [NEG_DI] Inverter for DINT values                       |
| [NEG_R] Inverter for REAL values                        |
| [SWICNV_B] Switching signal converter for Boolean       |
| [SWICNV_M] Switching signal converter for multistate    |
| [MRG_DA] Merge to date                                  |
| [MRG_DATI] Merge to date and time of day                |
| [MRGTIDAY] Merge to time of day                         |
| [TI_DI] Conversion time to double integer value         |
| [TI_DW] Converting time signal to Double Word           |
| [TIDAY] Time of day functions                           |
| [TIDAY_DW] Conversion time of day to double word value  |
| [DA] Date functions                                     |
| [DATI] Date and time of day functions                   |
| [DA_WORD] Conversion date to word value                 |
| [WORD_DA] Conversion word value to date                 |

# Function Blocks

| 多路选择器指令                                                                     |
|-----------------------------------------------------------------------------|
| [ENSEL_BO] Selection function Bool                                          |
| [ENSEL_MS] Selection function multistate                                    |
| [ENSEL_R] Selection function real                                           |
| [ENSEL_4V] Selection function for 4 values (2 multistate and 2 real values) |
| [SEL_BO] Multiplexer 1 of 2 for BOOL values                                 |
| [SEL_R] Multiplexer 1 of 2 for REAL values                                  |
| [SELBO_2M] Sel. Boolean f. 2 multistate                                     |
| [SELBO_BO] Selector Boolean for Boolean                                     |
| [SELBO_DI] Sel. Boolean f. Double Integer                                   |
| [SELBO_I] Selector Boolean for integer values                               |
| [SELBO_MS] Selector Boolean for multistate values                           |
| [SELBO_R] Selector Boolean for real values                                  |
| [SELBO_TI] Selector Boolean for time                                        |
| [SELMS_BO] Selector multistate for Boolean values                           |
| [SELMS_DI] MS selector for double integer value                             |
| [SELMS_I] Selector multistate for integer values                            |
| [SELMS_MS] Multistate selector for multistate                               |
| [SELMS_R] Selector multistate for real values                               |
| [SELMS_TI] Selector multistate for time                                     |

| 多路选择器指令                                                              |
|----------------------------------------------------------------------|
| [DMUX_101] Demultiplexer 101, 15 real and multistate and bool values |
| [DMUX8_BO] Demultiplexer 1 of 8 Boolean values                       |
| [DMUX8_I] Demultiplexer 1 of 8 integer values                        |
| [DMUX8_R] Demultiplexer 1 of 8 real values                           |
| [MUX_101] Multiplexer 101, 15 real and multistate and bool values    |
| [MUX_BO] Multiplexer 1 of n for BOOL values                          |
| [MUX_DI] Multiplexer 1 of n for DINT values                          |
| [MUX_I] Multiplexer 1 of n for INT values                            |
| [MUX2_R] Multiplexer 1 of 2 real values                              |
| [MUX4_R] Multiplexer 1 of 4 real values                              |
| [MUX8_MS] Multiplexer 1 of 8 for multistate                          |
| [MUX8_R] Multiplexer 1 of 8 real values                              |

# Function Blocks

| BACnet object 读指令                                  | BACnet object 写指令                                 |
|----------------------------------------------------|---------------------------------------------------|
| [R_A] Read analog                                  | [CMD_A] Command Analog                            |
| [R_AG] Read analog group                           | [CMD_AG] Command analog group                     |
| [R_ASCHED] Read analog scheduler                   | [CMD_B] Command binary                            |
| [R_B] Read binary                                  | [CMD_BG] Command binary group                     |
| [R_BG] Read binary group                           | [CMD_M] Command multistate                        |
| [R_BSCHED] Read binary scheduler                   | [CMD_MG] Command multistate group                 |
| [R_DA] Read date                                   | [W_A] Write analog                                |
| [R_DATI] Read date and time                        | [W_AG] Write analog group                         |
| [R_DINT] Read double integer value                 | [W_B] Write binary                                |
| [R_M] Read multistate                              | [W_BG] Write binary group                         |
| [R_MG] Read multistate group                       | [W_BO] WORD to BOOL conversion                    |
| [R_MSCHED] Read multistate scheduler               | [W_BY] WORD to BYTE conversion                    |
| [R_TI] Read time                                   | [W_DA] Write date                                 |
| [R_TIDAY] Read time of day                         | [W_DATI] Write date and time                      |
| [R_TRIG] Recognize positive edge and trigger pulse | [W_DINT] Write double integer value               |
| [R_UNSG] Read unsigned                             | [W_M] Write multistate                            |
| [RP_BOOL] Read property for Boolean value          | [W_MG] Write multistate group                     |
| [RP_DINT] Read property for double integer value   | [W_MPRVAL] Write multistate for present value     |
| [RP_REAL] Read property for real value             | [W_TI] Write time                                 |
| [RP_UNSG] Read property for unsigned value         | [W_TIDAY] Write time of day                       |
|                                                    | [W_UNSG] Write unsigned                           |
|                                                    | [WP_BOOL] Write property for Boolean value        |
|                                                    | [WP_DINT] Write property for double integer value |
|                                                    | [WP_REAL] Write property for real value           |
|                                                    | [WP_UNSG] Write property for unsigned value       |

# Function Blocks

## 回路控制指令

[CTR] Controller  
[CAS\_CTR] Cascade controller  
[CONT\_4ST] Step switch for 4 stages  
[SEQLINK] Sequence linker  
[SEQ\_DIR] Direct sequence linker  
[SEQ\_REV] Reverse sequence linker  
[SWI\_2P] Switch with adjustable switch-on/off point  
[SWI\_2PH] Two-position switch with switch-on point and hysteresis  
[FILTER] Attenuating filter  
[FILTER\_H] Hysteresis filter  
[CMDSEQ\_B] Command sequence for binary  
[ROT\_8] Rotation switch with 8 outputs  
[SELP\_8MS] Profile and table value selection for 8 multistate values

## HVAC 相关指令

[HCRV] Heating curve  
[HRA] Heating release algorithm  
[PRDVHCTL] Predictive heating controller  
[PURGE] Purge optimization  
[NGTC] Night cooling  
[OSSC] Optimum start stop control  
[SOL\_FCD] Solar function facade  
[SOL\_POS] Solar position calculation  
[SP\_SPLY] Determination of setpoint supply air  
[MON\_ERC] Energy recovery monitoring  
[HX\_TPHI] Enthalpy and absolute humidity to temperature and relative humidity  
[TPHI\_HX] Temperature and relative humidity to enthalpy and absolute humidity  
[LOCK\_LM] Locking limitation

## 接口指令

[PAR\_BO] Boolean parameter  
[PAR\_I] Integer parameter  
[PAR\_R] Real parameter  
[PAR\_T] Time parameter  
[PAR\_DATI] Date and time of day parameter

## 系统指令

[AS\_STTUP] Automation station start-up information  
[CMN\_EVT] Common event  
[CommonDB] Common Data-Block  
EN and ENO parameters  
[GET\_DATI] Get local date and time of day  
[KICKFNCT] Kick function  
[PRIO\_CNV] Priority converter to 4 Boolean values  
[RST\_CNT] Counter restart